

Final Project Calculation of Lake Pollutants

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Problem 1

The initial conditions are $C_1(0) = C_{10}$ and $C_2(0) = C_{20}$. Lake L_1 is affected by pollutant inflow and outflow. We assume that water with a flow rate of $Q \text{ m}^3/\text{day}$ (with a pollutant concentration of $C_0 \text{ g/m}^3$) flows into Lake L_1 , and at the same time, water with the same flow rate $Q \text{ m}^3/\text{day}$ flows out of Lake L_1 into Lake L_2 . So that

$$\frac{d(V_1 C_1(t))}{dt} = QC_0(t) - QC_1(t)$$

Similarly, the equation plus external pollutant input, i.e., pollutants with a mass of $W \text{ g}$ discharged into Lake L_2 per day (unit: g/day) will be

$$\frac{d(V_2 C_2(t))}{dt} = QC_1(t) - QC_2(t) + W$$

Since $\beta_1 = \frac{Q}{V_1}$, $\beta_2 = \frac{Q}{V_2}$ and $\beta_3 = \frac{W}{V_2}$, we divide the equations by V_1 and V_2 respectively and then have

$$\begin{cases} \frac{dC_1(t)}{dt} = \beta_1 C_0(t) - \beta_1 C_1(t), \\ \frac{dC_2(t)}{dt} = \beta_2 C_1(t) - \beta_2 C_2(t) + \beta_3, \\ C_1(0) = C_{10}, \\ C_2(0) = C_{20}. \end{cases}$$

Problem 2

The solutions are:

$$C_1(t) = C_0 - C_0 \cdot \exp(-\beta_1 t) + C_{10} \cdot \exp(-\beta_1 t)$$

$$C_2(t) = \exp(-\beta_1 t) \left(\frac{\beta_2(C_0 - C_{10})}{\beta_1 - \beta_2} - \frac{C_0 \beta_2 \exp(\beta_1 t)}{\beta_1 - \beta_2} \right) - \exp(-\beta_2 t) \left(\frac{\beta_1 \beta_3 - \beta_2 \beta_3 - C_{10} \beta_2^2 + C_{20} \beta_2^2 + C_0 \beta_1 \beta_2 - C_{20} \beta_1 \beta_2}{\beta_2(\beta_1 - \beta_2)} - \frac{\exp(\beta_2 t)(C_0 \beta_1 - \beta_3 + \frac{\beta_1 \beta_3}{\beta_2})}{\beta_1 - \beta_2} \right)$$

```

1  %-----Problem 2-----
2  clc; clear; close all;
3
4  % Define Variables
5  syms C1(t) C2(t) beta1 beta2 beta3 C0 C10 C20
6
7  % Define Equations
8  eq1 = diff(C1, t) == beta1 * (C0 - C1);
9  eq2 = diff(C2, t) == beta2 * (C1 - C2) + beta3;
10
11 % Define Initial Conditions
12 cond1 = C1(0) == C10;
13 cond2 = C2(0) == C20;
14
15 % Solve the ODE System
16 [C1_sol, C2_sol] = dsolve([eq1, eq2], [cond1, cond2]);
17
18 % Simplify the Solutions
19 C1_sol = simplify(C1_sol);
20 C2_sol = simplify(C2_sol);
21
22 % Display the Analytical Solutions
23 disp('C1(t) = ');
24 disp(C1_sol);
25 disp('C2(t) = ');
26 disp(C2_sol);
27
28 % Output:
29 % C1(t) =
30 % C0 - C0*exp(-beta1*t) + C10*exp(-beta1*t)
31 % C2(t) =
32 % exp(-beta1*t)*((beta2*(C0 - C10))/(beta1 - beta2) - ...
    (C0*beta2*exp(beta1*t))/(beta1 - beta2)) - ...
    exp(-beta2*t)*((beta1*beta3 - beta2*beta3 - C10*beta2^2 + ...
    C20*beta2^2 + C0*beta1*beta2 - C20*beta1*beta2)/(beta2*(beta1 ...
    - beta2)) - (exp(beta2*t)*(C0*beta1 - beta3 + ...
    (beta1*beta3)/beta2))/(beta1 - beta2))

```

Problem 3

We use the analytical solution to fit the model. Since there's only 1 parameter β_1 in the first equation, we adopt the two steps `lsqcurvefit`. The capacity of lake L_1 is estimated to be 200000.00 m³ and the capacity of lake L_2 is estimated to be 100000.00 m³.

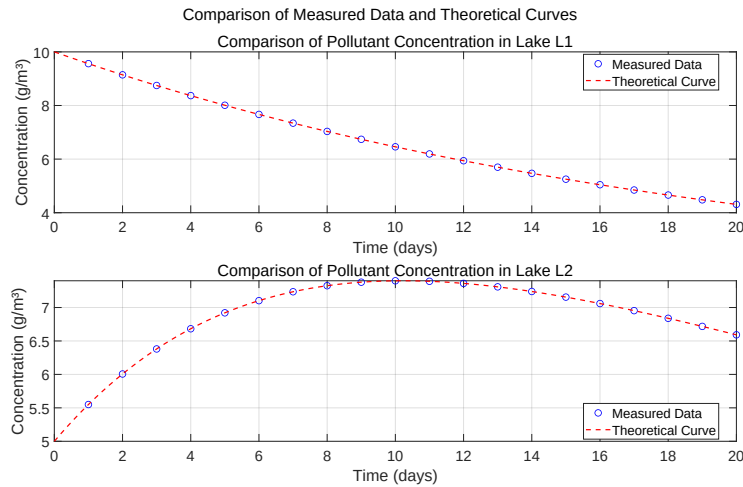


Figure 1: Problem 3

```

1  %-----Problem 3-----
2  clc; clear; close all;
3
4  % Data and Parameters
5  data = readmatrix('data.xls', 'NumHeaderLines', 1);
6  t_data = data(:, 1);
7  C1_data = data(:, 2);
8  C2_data = data(:, 3);
9  Q = 1e4; % m³/day
10 C0 = 1; % g/m³
11 C10 = 10; % g/m³
12 C20 = 5; % g/m³
13 W = 1e4; % g/day
14 K = C0 + W/Q; % = 2 g/m³
15
16 %% Estimate V Sequentially
17 % Define V1 Model Function
18 model_C1 = @(beta1, t) C0 + (C10 - C0) * exp(-beta1 * t);
19 beta1_init = 0.1; % Initial Guess Values
20 opts = optimoptions('lsqcurvefit', 'Display', 'off');
21 beta1_fit = lsqcurvefit(@(b,t) model_C1(b,t), beta1_init, t_data, ...
22     C1_data, [], [], opts);
23 V1 = Q / beta1_fit;

```

```

23
24 % Define V2 Model Function
25 model_C2 = @(beta2, t) ...
26     (C20 - K - (beta2 * (C10 - C0)) / (beta2 - beta1_fit)) .* ...
27     exp(-beta2 * t) + ...
28     K + (beta2 * (C10 - C0) / (beta2 - beta1_fit)) .* ...
29     exp(-beta1_fit * t);
30 beta2_init = 0.05; % Initial Guess Values
31 beta2_fit = lsqcurvefit(model_C2, beta2_init, t_data, C2_data, ...
32     [], [], opts);
33 V2 = Q / beta2_fit;
34
35 %% Display Estimated Results
36 fprintf('Estimated lake capacities:\n');
37 fprintf('V1 = %.2f m^3\n', V1);
38 fprintf('V2 = %.2f m^3\n', V2);
39
40
41 t_fit = linspace(0, max(t_data), 100);
42 C1_fit = model_C1(beta1_fit, t_fit);
43 C2_fit = model_C2(beta2_fit, t_fit);
44
45 %% Plot and Export
46 figure;
47 subplot(2, 1, 1);
48 plot(t_data, C1_data, 'bo', 'LineWidth', 1, 'MarkerSize', 8, ...
49     'DisplayName', 'Measured Data');
50 hold on;
51 plot(t_fit, C1_fit, 'r—', 'LineWidth', 1.5, 'DisplayName', ...
52     'Theoretical Curve');
53 title('Comparison of Pollutant Concentration in Lake L1', ...
54     'FontSize', 12);
55 xlabel('Time (days)', 'FontSize', 10);
56 ylabel('Concentration (g/m^3)', 'FontSize', 10);
57 legend('Location', 'best');
58 set(gca, 'FontSize', 18);
59 grid on;
60
61 subplot(2, 1, 2);
62 plot(t_data, C2_data, 'bo', 'LineWidth', 1, 'MarkerSize', 8, ...
63     'DisplayName', 'Measured Data');
64 hold on;
65 plot(t_fit, C2_fit, 'r—', 'LineWidth', 1.5, 'DisplayName', ...
66     'Theoretical Curve');
67 title('Comparison of Pollutant Concentration in Lake L2', ...
68     'FontSize', 12);
69 xlabel('Time (days)', 'FontSize', 10);
70 ylabel('Concentration (g/m^3)', 'FontSize', 10);
71 legend('Location', 'best');
72 set(gca, 'FontSize', 18);
73 grid on;
74
75 sgtitle('Comparison of Measured Data and Theoretical Curves', ...
76     'FontSize', 20);
77 set(gcf, 'Position', [100, 100, 1600, 900]);
78 exportgraphics(gcf, 'Problem3.pdf', 'ContentType', 'vector');
79
80 % Output:

```

```

70 % Estimated lake capacities:
71 % V1 = 200000.00 m^3
72 % V2 = 100000.00 m^3

```

Problem 4

It seems that it takes 245 days for the concentration of pollutants in lake L_1 and lake L_2 to both be below 2 g/m^3 by calculation. However, we can see from the figure that $C_2(t)$ can never be below 2 g/m^3 . So the required time calculated is due to error. So the time when both are below 2 g/m^3 can never be achieved.

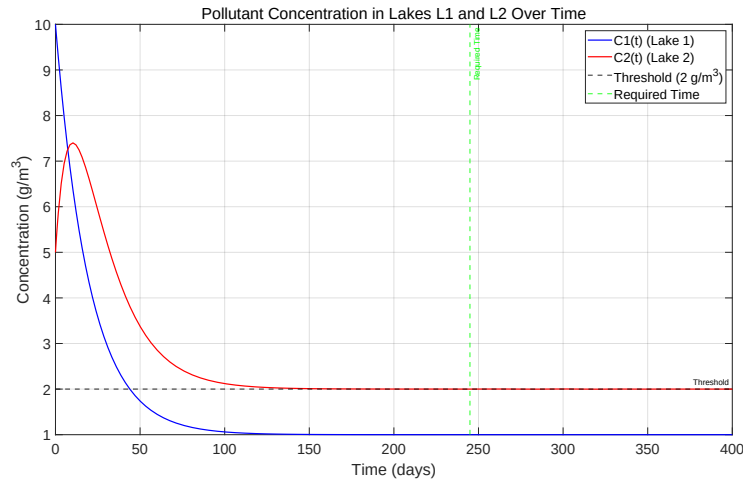


Figure 2: Problem 4

```

1 %-----Problem 4-----
2 clc; clear; close all;
3
4 %% Parameters and Coefficients
5 Q = 1e4; % m^3/day
6 C0 = 1; % g/m^3
7 C10 = 10; % g/m^3
8 C20 = 5; % g/m^3
9 W = 1e4; % g/day
10 V1 = 200000.00; % m^3
11 V2 = 100000.00; % m^3
12
13 beta1 = Q / V1;
14 beta2 = Q / V2;

```

```

15  beta3 = W / V2;
16
17  %% Solve ODE
18  odefun = @(t, C) [beta1 * (C0 - C(1)); beta2 * (C(1) - C(2)) + ...
    beta3];
19
20  t_span = [0 400];
21  [t, C] = ode45(odefun, t_span, [C10; C20]);
22
23  % Find the Time Below 2 g/m^3
24  idx = find(C(:, 1) < 2 & C(:, 2) < 2, 1, 'first');
25  if ~isempty(idx)
26      time_required = t(idx);
27      fprintf('It takes %.2f days for C1 and C2 to both be below 2 ...
    g/m^3\n', time_required);
28  else
29      fprintf('The requirement is not met within 400 days\n');
30  end
31
32  %% Plotting and Save
33  figure;
34  plot(t, C(:, 1), 'b-', 'LineWidth', 1.5);
35  hold on;
36  plot(t, C(:, 2), 'r-', 'LineWidth', 1.5);
37  yline(2, 'k—', 'Threshold', 'LineWidth', 1.5);
38  if ~isempty(idx)
39      xline(time_required, 'g—', 'Required Time', 'LineWidth', 1.5);
40  end
41  xlabel('Time (days)');
42  ylabel('Concentration (g/m^3)');
43  title('Pollutant Concentration in Lakes L1 and L2 Over Time');
44  legend('C1(t) (Lake 1)', 'C2(t) (Lake 2)', 'Threshold (2 g/m^3)', ...
    'Required Time');
45  grid on;
46
47  set(gca, 'FontSize', 18);
48  set(gcf, 'Position', [100, 100, 1600, 900]);
49  exportgraphics(gcf, 'Problem4.pdf', 'ContentType', 'vector');
50
51  % Output:
52  % It takes 244.95 days for C1 and C2 to both be below 2 g/m^3

```

Problem 5

The target cannot be achieved within 40 days even for the maximum 15000.00 m³/day. For $Q = 15000.00$ m³/day, the target is achieved after 54 days.

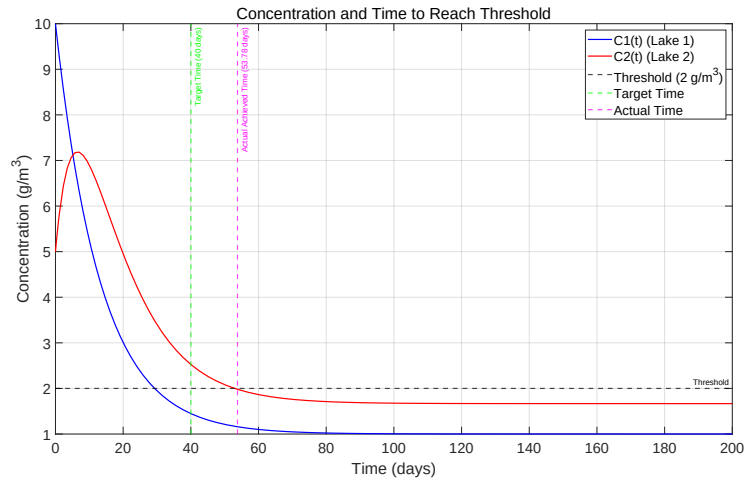


Figure 3: Problem 5

```

1  %-----Problem 5-----
2  clc; clear; close all;
3
4  %% Parameters
5  C0 = 1;      % g/m^3
6  C10 = 10;    % g/m^3
7  C20 = 5;     % g/m^3
8  W = 1e4;     % g/day
9  V1 = 200000.00; % m^3
10 V2 = 100000.00; % m^3
11 Q_max = 1.5e4; % m^3/day
12 t_target = 40; % days
13
14 %% Search for Q and Solve ODE
15 Q_values = linspace(0, Q_max, 100);
16 found = false;
17 Q_opt = Q_max;
18 t_required = NaN;
19
20 for Q = Q_values
21     beta1 = Q / V1;
22     beta2 = Q / V2;
23     beta3 = W / V2;
24     odefun = @(t, C) [beta1 * (C0 - C(1)); beta2 * (C(1) - C(2)) ...
25                       + beta3];
26     [t, C] = ode45(odefun, [0 t_target], [C10; C20]);
27     if C(end,1) < 2 && C(end,2) < 2
28         fprintf('For Q = %.2f m^3/day, the target is achieved ...
29                 within 40 days\n', Q);
30         Q_opt = Q;
31         t_required = t(end);
32         found = true;
33         break

```

```

32     end
33 end
34
35 if ~found
36     fprintf('The target cannot be achieved within 40 days for ...
37         Q_max = %.2f m^3/day\n', Q_max);
38     Q_opt = Q_max;
39     beta1 = Q_opt / V1;
40     beta2 = Q_opt / V2;
41     beta3 = W / V2;
42     odefun = @(t, C) [beta1 * (C0 - C(1)); beta2 * (C(1) - C(2)) ...
43         + beta3];
44     [t_long, C_long] = ode45(odefun, [0 200], [C10; C20]);
45
46     idx = find(C_long(:,1) < 2 & C_long(:,2) < 2, 1);
47     if isempty(idx)
48         fprintf('Even after 200 days, concentrations do not fall ...
49             below 2 g/m^3.\n');
50         t_required = NaN;
51     else
52         t_required = t_long(idx);
53         fprintf('For Q = %.2f m^3/day, the target is achieved ...
54             after %.2f days.\n', Q_opt, t_required);
55     end
56
57     t = t_long;
58     C = C_long;
59 end
60
61 %% Plotting and Save
62 figure;
63 plot(t, C(:, 1), 'b-', 'LineWidth', 1.5); hold on;
64 plot(t, C(:, 2), 'r-', 'LineWidth', 1.5);
65 yline(2, 'k-', 'Threshold', 'LineWidth', 1.5);
66 xline(t_target, 'g-', 'Target Time (40 days)', 'LineWidth', 1.5);
67 if ~isnan(t_required)
68     xline(t_required, 'm-', sprintf('Actual Achieved Time (%.2f ...
69         days)', t_required), 'LineWidth', 1.5);
70 end
71
72 title('Concentration and Time to Reach Threshold');
73 xlabel('Time (days)');
74 ylabel('Concentration (g/m^3)');
75 legend('C1(t) (Lake 1)', 'C2(t) (Lake 2)', ...
76     'Threshold (2 g/m^3)', 'Target Time', 'Actual Time');
77 grid on;
78
79 set(gca, 'FontSize', 18);
80 set(gcf, 'Position', [100, 100, 1600, 900]);
81 exportgraphics(gcf, 'Problem5.pdf', 'ContentType', 'vector');
82
83 % Output:
84 % The target cannot be achieved within 40 days for Q_max = ...
85 %     15000.00 m^3/day
86 % For Q = 15000.00 m^3/day, the target is achieved after 53.78 days.

```


Problem 6

First we use **moving average filtering** to process the data. Then we use **lsqcurvefit** the same way as before. Flow rate Q is estimated to be 14818.00 m^3/day . The capacity of lake L_1 is estimated to be 296626.84 m^3 and the capacity of lake L_2 is estimated to be 196002.95 m^3 . The answer might be wrong, since the figure and the data are much the same as in the problems before, but the results turn out to be different. I doublecheck the value estimated in step 1 and find that $\beta_1 = 0.0500$ is the same, which indicates that the data smoothing process is correct. While in step 2, $\beta_2 = 0.0756$. If it is not a problem of data modification, it must be due to the introduction of the other parameter Q . It may be unreasonable to estimate two parameter just given 1 nonlinear equation.

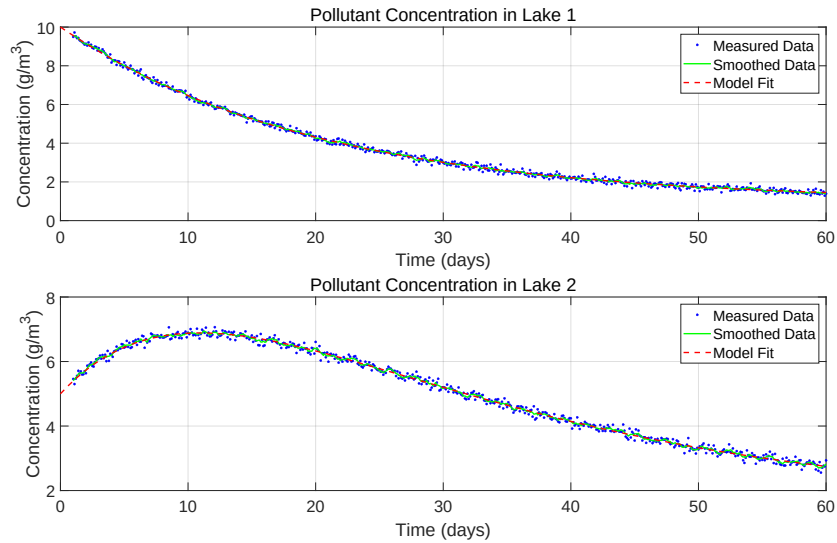


Figure 4: Problem 6

```

1  %-----Problem 6-----
2  clc; clear; close all;
3
4  %% Parameters
5  data = readmatrix('noisydata.xls', 'NumHeaderLines', 1);
6  t_data = data(:, 1);
7  C1_data = data(:, 2);
8  C2_data = data(:, 3);
9  C0 = 1;      % g/m³
10 C10 = 10;    % g/m³
11 C20 = 5;     % g/m³
12 W = 1e4;    % g/day
13

```

```

14 %% Data Processing
15 window_size = 5;
16 C1_smooth = movmean(C1_data, window_size);
17 C2_smooth = movmean(C2_data, window_size);
18
19 %% Estimate V Sequentially
20 % Define V1 Model Function
21 model_C1 = @(beta1, t) C0 + (C10 - C0) * exp(-beta1 * t);
22 beta1_init = 0.05; % Initial Guess Values
23 opts = optimoptions('lsqcurvefit', 'Display', 'off');
24 beta1_fit = lsqcurvefit(@(b,t) model_C1(b,t), beta1_init, t_data, ...
    C1_smooth, [], [], opts);
25
26 % Define V2 Model Function
27 model_C2 = @(para, t) ...
28     (C20 - (C0 + W/para(2)) - (para(1) * (C10 - C0)) / (para(1) - ...
    beta1_fit)) .* exp(-para(1) * t) + ...
29     C0 + W/para(2) + (para(1) * (C10 - C0)) / (para(1) - ...
    beta1_fit)) .* exp(-beta1_fit * t);
30 para_init = [0.1, 1e4]; % Initial Guess Values
31 para_fit = lsqcurvefit(model_C2, para_init, t_data, C2_smooth, ...
    [], [], opts);
32 V1 = para_fit(2) / beta1_fit;
33 V2 = para_fit(2) / para_fit(1);
34
35 % Display Results
36 fprintf('Estimated Q = %.2f m^3/day\n', para_fit(2));
37 fprintf('Estimated V1 = %.2f m^3\n', V1);
38 fprintf('Estimated V2 = %.2f m^3\n', V2);
39
40 % Calculate Model Predictions
41 t_fit = linspace(0, max(t_data), 100);
42 C1_fit = model_C1(beta1_fit, t_fit);
43 C2_fit = model_C2(para_fit, t_fit);
44
45 %% Plotting and Saving
46 figure;
47 subplot(2,1,1);
48 plot(t_data, C1_data, 'b.', 'MarkerSize', 8);
49 hold on;
50 plot(t_data, C1_smooth, 'g-', 'LineWidth', 1.5);
51 plot(t_fit, C1_fit, 'r-', 'LineWidth', 1.5);
52 title('Pollutant Concentration in Lake 1');
53 xlabel('Time (days)');
54 ylabel('Concentration (g/m^3)');
55 legend('Measured Data', 'Smoothed Data', 'Model Fit');
56 set(gca, 'FontSize', 18);
57 grid on;
58
59 subplot(2,1,2);
60 plot(t_data, C2_data, 'b.', 'MarkerSize', 8);
61 hold on;
62 plot(t_data, C2_smooth, 'g-', 'LineWidth', 1.5);
63 plot(t_fit, C2_fit, 'r-', 'LineWidth', 1.5);
64 title('Pollutant Concentration in Lake 2');
65 xlabel('Time (days)');
66 ylabel('Concentration (g/m^3)');

```

```

67 legend('Measured Data', 'Smoothed Data', 'Model Fit');
68 set(gca, 'FontSize', 18);
69 grid on;
70
71 set(gcf, 'Position', [100, 100, 1600, 900]);
72 exportgraphics(gcf, 'Problem6.pdf', 'ContentType', 'vector');
73
74 % Output:
75 % Estimated Q = 14818.00 m3/day
76 % Estimated V1 = 296626.84 m3
77 % Estimated V2 = 196002.95 m3

```